SECURITY MONITORING USING WAZUH SIEM

Submitted by

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1. ABSTRACT

This project demonstrates the implementation of a basic Security Operations Center (SOC) monitoring environment using the Wazuh SIEM platform. The system consists of three virtual machines: a Kali Linux machine used as the attacker, an Ubuntu system acting as the victim machine, and a Wazuh server responsible for monitoring and analyzing security events. The Wazuh agent was installed on the victim machine to collect system logs and forward them to the Wazuh server. A brute-force attack on the SSH service was simulated using the Hydra tool from the Kali Linux machine. The generated authentication events were detected and displayed in the Wazuh dashboard, demonstrating how SIEM platforms can be used for security monitoring and log analysis. This project demonstrates how a SIEM system like Wazuh can detect suspicious login attempts and help security analysts monitor systems in a SOC environment.

2. OBJECTIVE

- To create a basic SOC monitoring lab using Wazuh SIEM.
- To collect system logs from a victim machine using the Wazuh agent.
- To simulate an SSH brute-force attack using Hydra.
- To analyze the generated alerts in the Wazuh dashboard.

3. TOOLS AND TECHNOLOGIES USED

Tool	Purpose
Kali Linux	Attack machine
Ubuntu	Victim machine
Wazuh	SIEM monitoring platform
VirtualBox	Virtual environment
Hydra	Brute-force attack tool

4. SYSTEM SETUP

The project environment consists of three virtual machines connected through a virtual network.

Kali Linux – Attacker – 192.168.25.99

Wazuh Server – Monitoring Server – 192.168.25.100

Ubuntu – Victim Machine – 192.168.25.59

The victim machine runs the Wazuh agent which sends system logs to the Wazuh server for analysis.

5. SYSTEM ARCHITECTURE

Kali Linux (Attacker) 192.168.25.99

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SSH brute-force attack

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Victim Machine 192.168.25.59 (Wazuh Agent)

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Log data

▼

Wazuh Server 192.168.25.100

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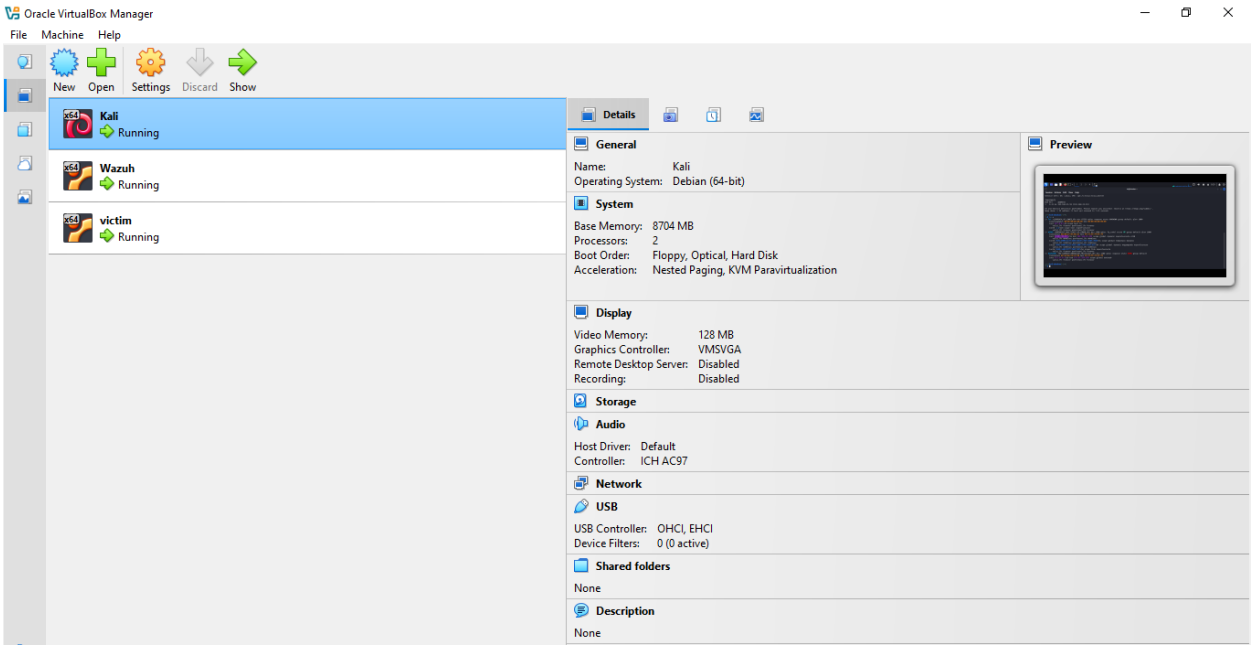
Wazuh Dashboard (Security Monitoring)

Figure 1: VirtualBox showing 3 VMs running



6. IMPLEMENTATION STEPS

1. Installed VirtualBox to create the virtual environment.

2. Installed Kali Linux virtual machine for attack simulation.

3. Installed Ubuntu Linux as the victim machine.

4. Installed the Wazuh server for monitoring security events.

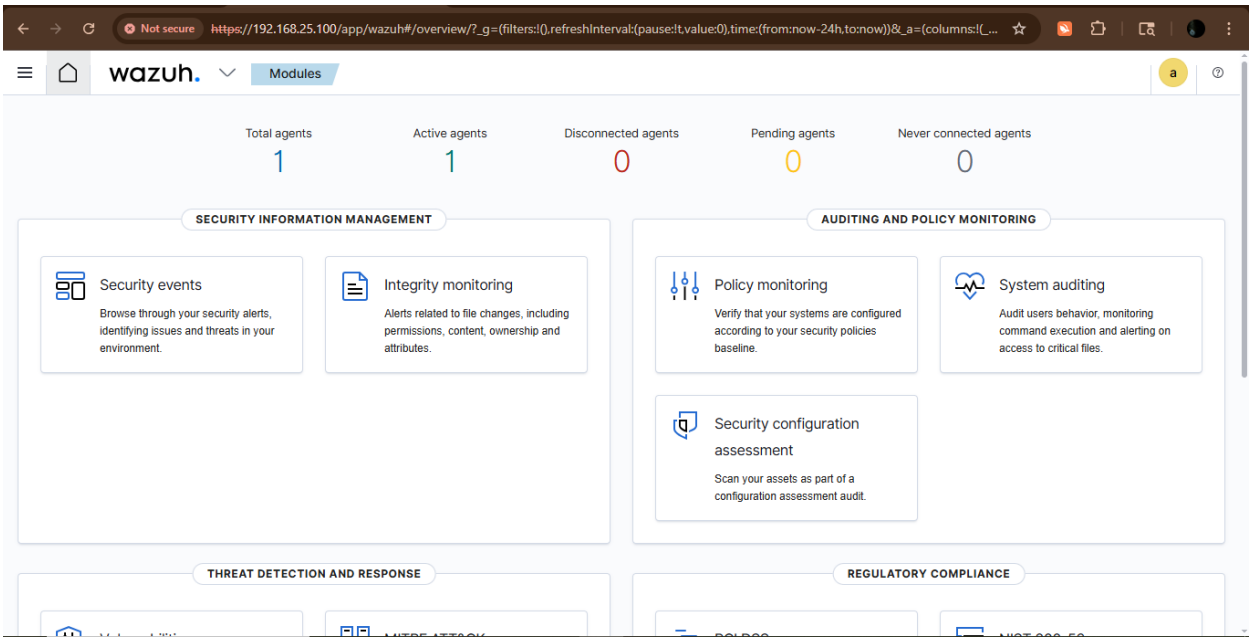
5. Installed the Wazuh agent on the victim machine.

6. Connected the victim agent to the Wazuh manager.

7. Verified the agent status in the Wazuh dashboard.

8. Monitored system logs and alerts through the Wazuh interface.

Figure 2: Wazuh dashboard → Agent Active



7. ATTACK SIMULATION – SSH BRUTE FORCE ATTACK

A brute-force attack was simulated using Hydra to perform multiple login attempts on the SSH service of the victim machine.

Command used:

hydra -l root -P /usr/share/wordlists/rockyou.txt 192.168.25.59 ssh

Result:

Multiple SSH login attempts were generated.

Authentication events were recorded on the victim system.

Wazuh detected and displayed the events in the dashboard.

Figure 3: Hydra SSH Brute Force Attack from Kali Linux

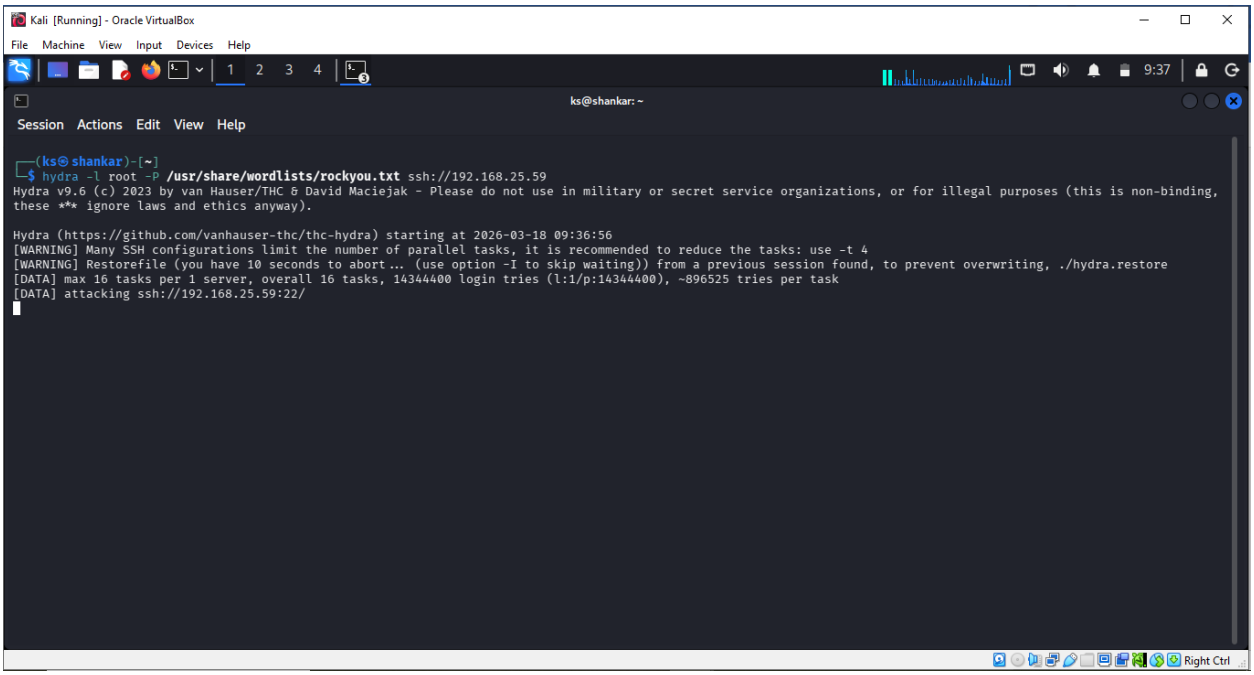
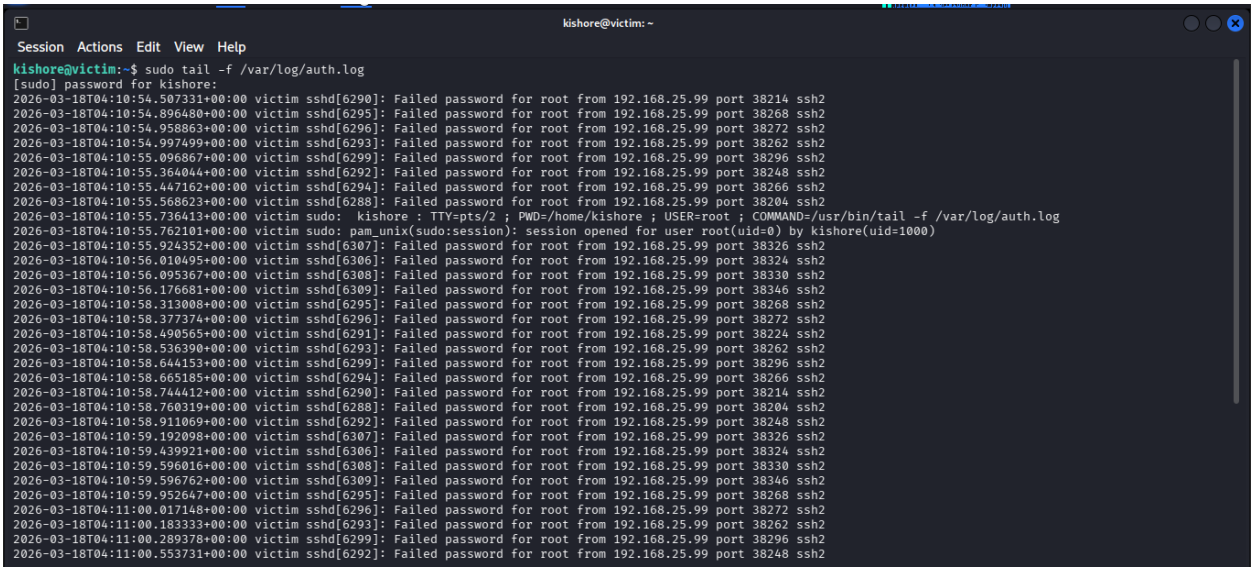


Figure 4: Victim Machine Logs / Authentication Attempts (/var/log/auth.log)



8. MONITORING AND ALERTS

The Wazuh dashboard displayed security events related to the attack including:

PAM login session opened

PAM login session closed

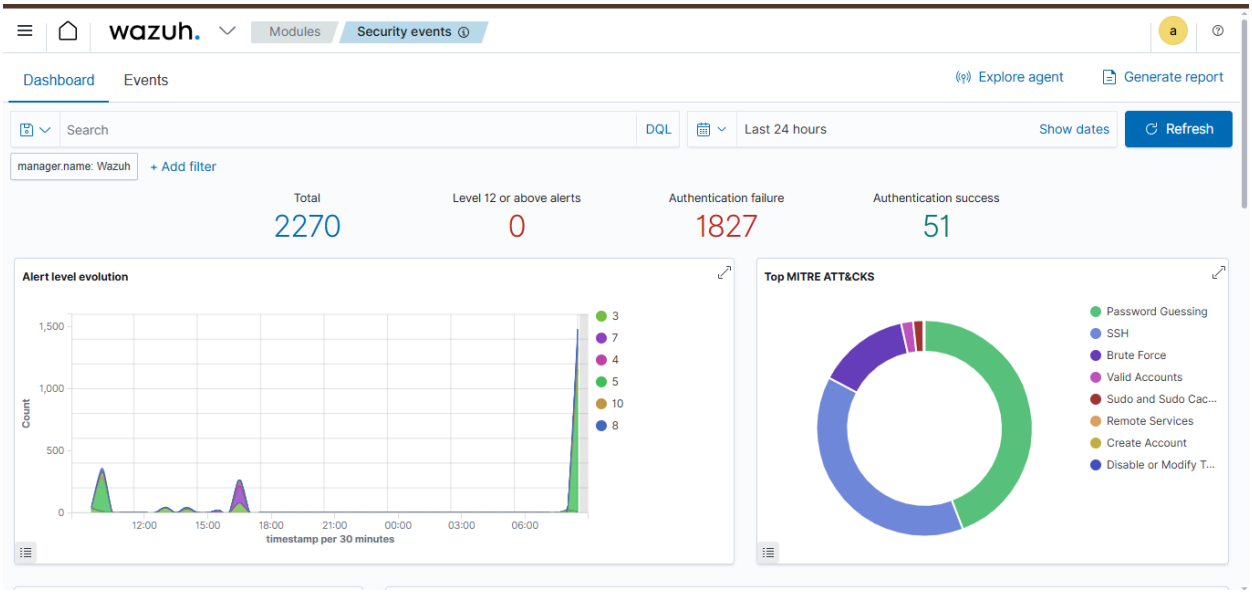
Authentication related events

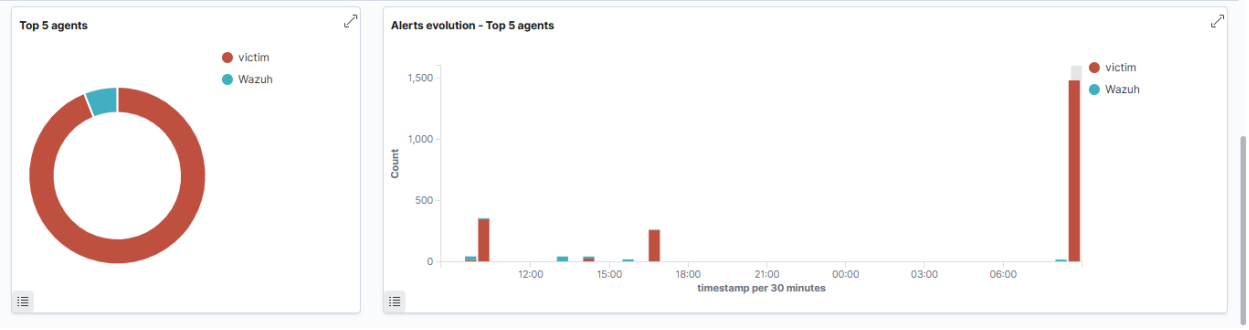
Privilege escalation alerts

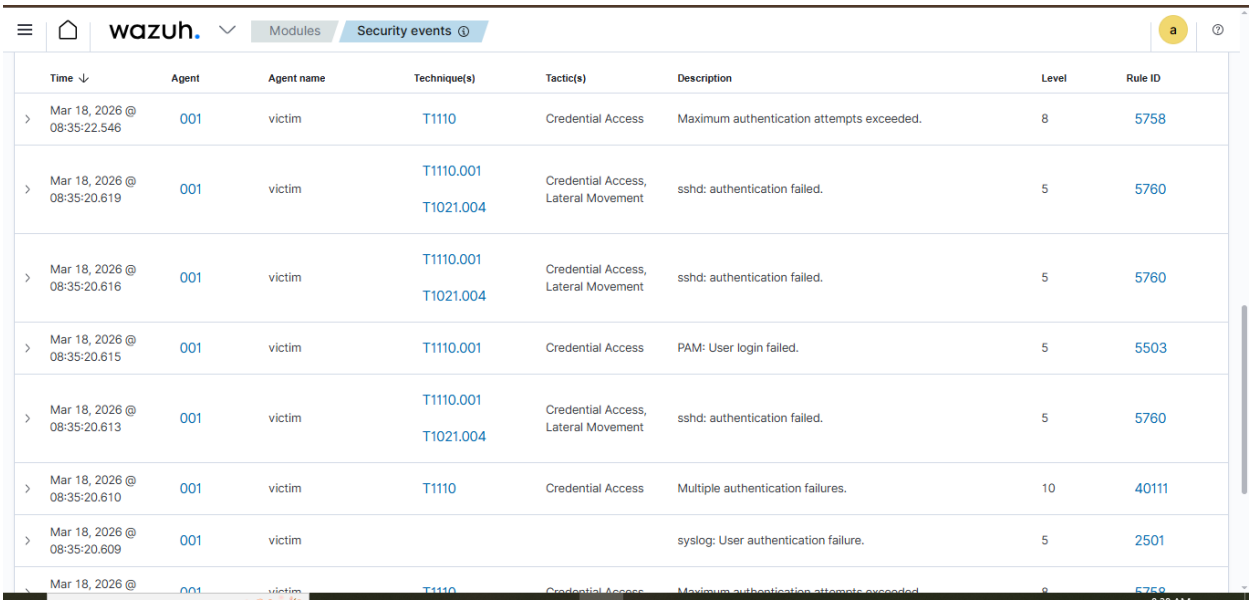
Rootcheck anomaly detection

These alerts confirmed that the Wazuh monitoring system successfully detected activities occurring on the victim machine.

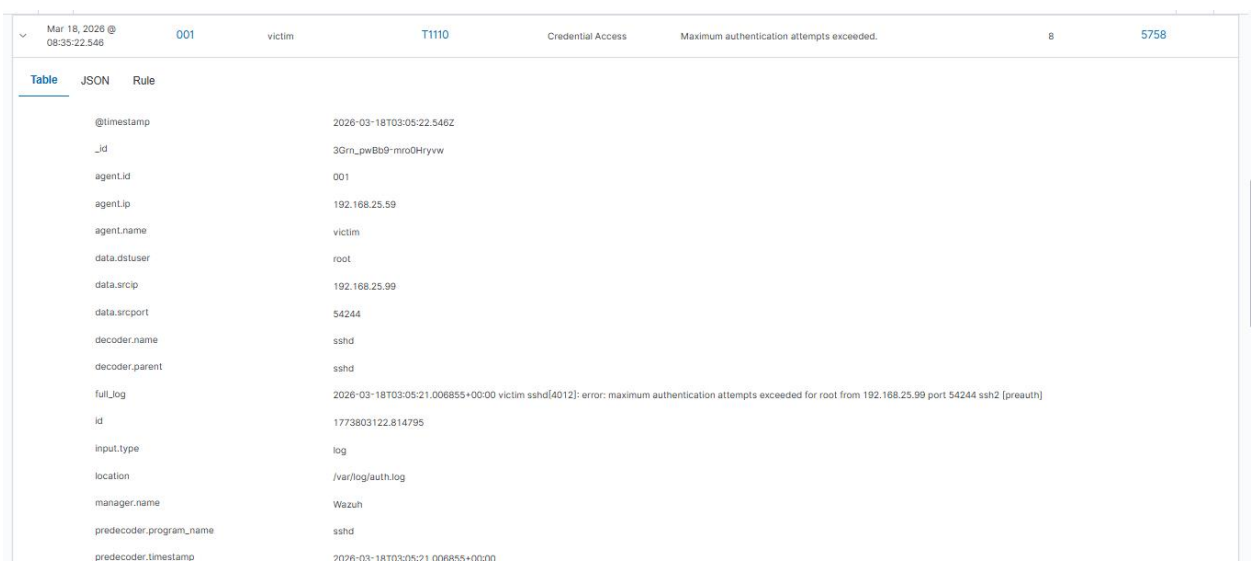
Figure 5: Wazuh Security Events Dashboard







Time	Agent	Agent name	Technique(s)	Tactic(s)	Description	Level	Rule ID
Mar 18, 2026 @ 08:35:22.546	001	victim	T1110	Credential Access	Maximum authentication attempts exceeded.	8	5758
Mar 18, 2026 @ 08:35:20.619	001	victim	T1110.001 T1021.004	Credential Access, Lateral Movement	sshd: authentication failed.	5	5760
Mar 18, 2026 @ 08:35:20.616	001	victim	T1110.001 T1021.004	Credential Access, Lateral Movement	sshd: authentication failed.	5	5760
Mar 18, 2026 @ 08:35:20.615	001	victim	T1110.001	Credential Access	PAM: User login failed.	5	5503
Mar 18, 2026 @ 08:35:20.613	001	victim	T1110.001 T1021.004	Credential Access, Lateral Movement	sshd: authentication failed.	5	5760
Mar 18, 2026 @ 08:35:20.610	001	victim	T1110	Credential Access	Multiple authentication failures.	10	40111
Mar 18, 2026 @ 08:35:20.609	001	victim			syslog: User authentication failure.	5	2501
Mar 18, 2026 @	001	victim	T1110	Credential Access	Maximum authentication attempts exceeded.	8	5758



9. LOG ANALYSIS

In this project, log analysis was performed using the **Wazuh Security Events dashboard** to identify suspicious activities generated during the SSH brute-force attack.

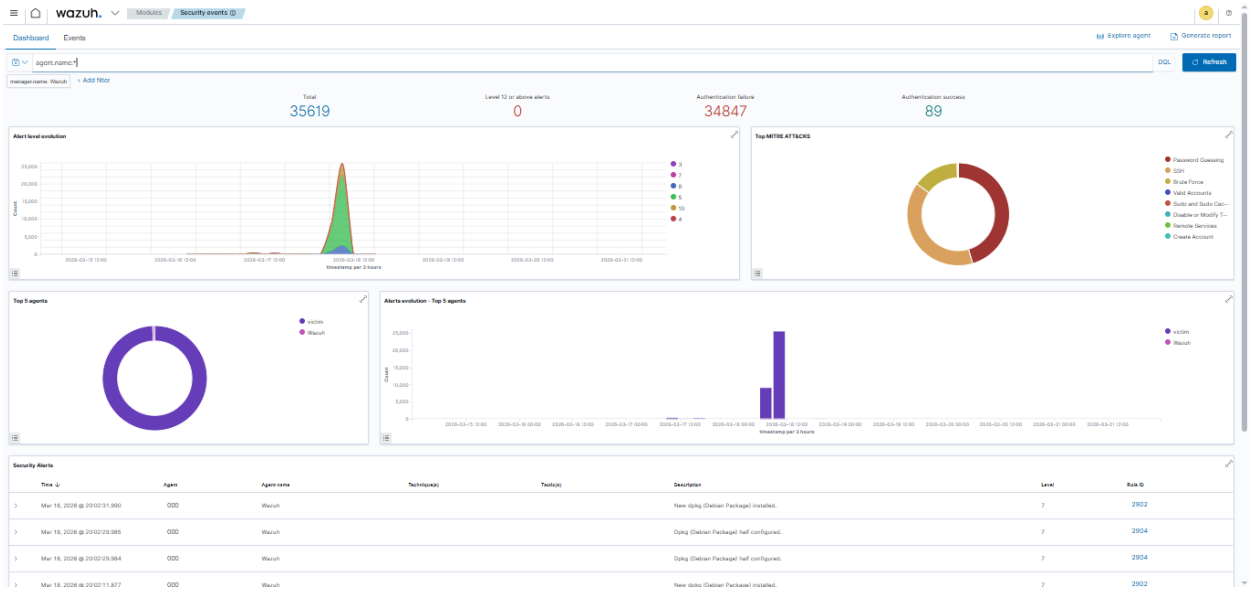
The Wazuh agent installed on the victim machine collected system logs and forwarded them to the Wazuh server. These logs were then analyzed using different search queries in the dashboard.

The following queries were used for log analysis:

1.agent.name:*

This query was used to display all logs generated by all connected agents in the Wazuh system. It helps verify that the victim machine is actively sending logs to the Wazuh server.

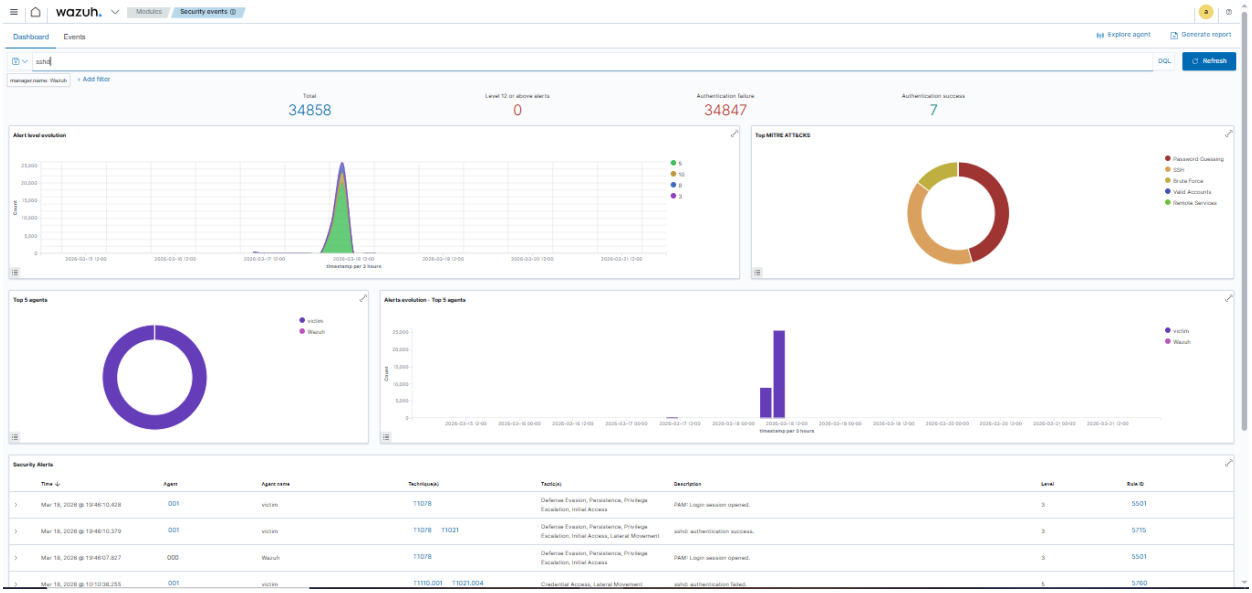
Figure 6: Wazuh Dashboard – All Agent Logs (agent.name:*)



2.sshd

This query filters logs related to the SSH service. It shows authentication attempts, login sessions, and other SSH-related events generated during the brute-force attack.

Figure 7: SSH Authentication Events (sshd)



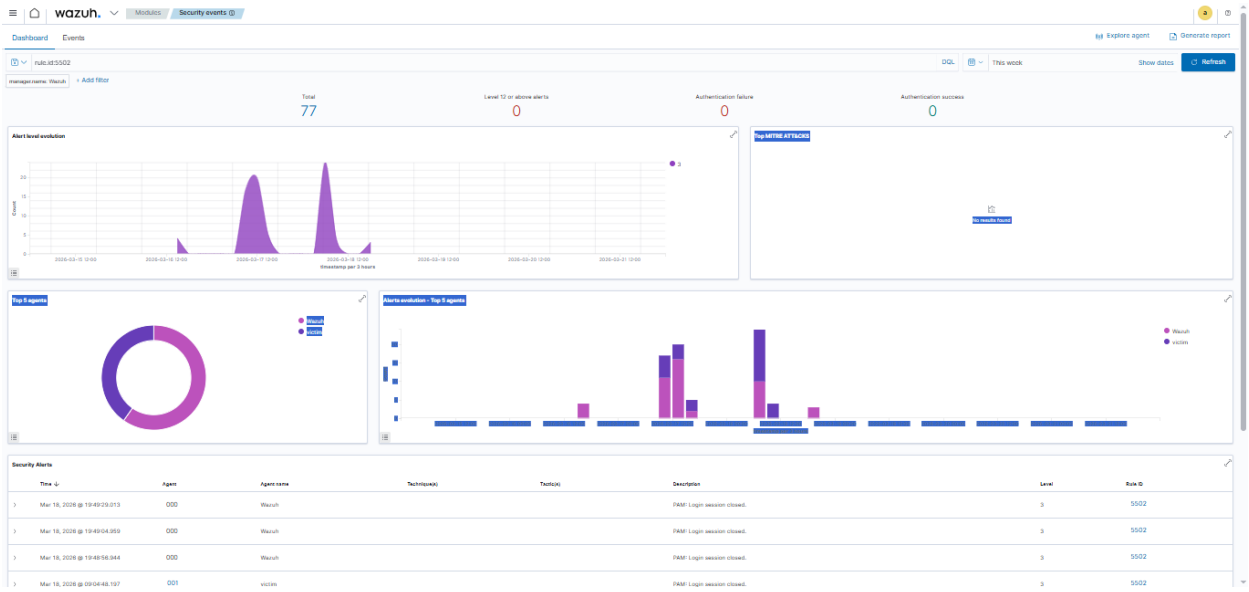
3.rule.id:5501

This query was used to identify specific security alerts related to **PAM login session opened events**. These alerts indicate successful login activity on the monitored system.

Using these queries, the Wazuh dashboard displayed authentication failures, successful login attempts, and security alerts generated during the simulated attack.

The log analysis confirms that the Wazuh SIEM platform successfully detected and monitored the SSH attack activities on the victim machine.

Figure 8: Security Alerts Using Rule ID 5501 (rule.id:5501)



10. RESULT

The SOC monitoring environment successfully collected and analyzed logs from the victim machine using the Wazuh SIEM platform. The simulated SSH brute-force attack generated authentication events that were monitored and displayed in the Wazuh dashboard.

11. CONCLUSION

The project successfully demonstrated how a SIEM system can monitor system activity and detect suspicious behavior. By simulating an SSH brute-force attack, we observed how Wazuh collects logs and generates alerts. This project provided practical exposure to SOC operations, log analysis, and security monitoring.